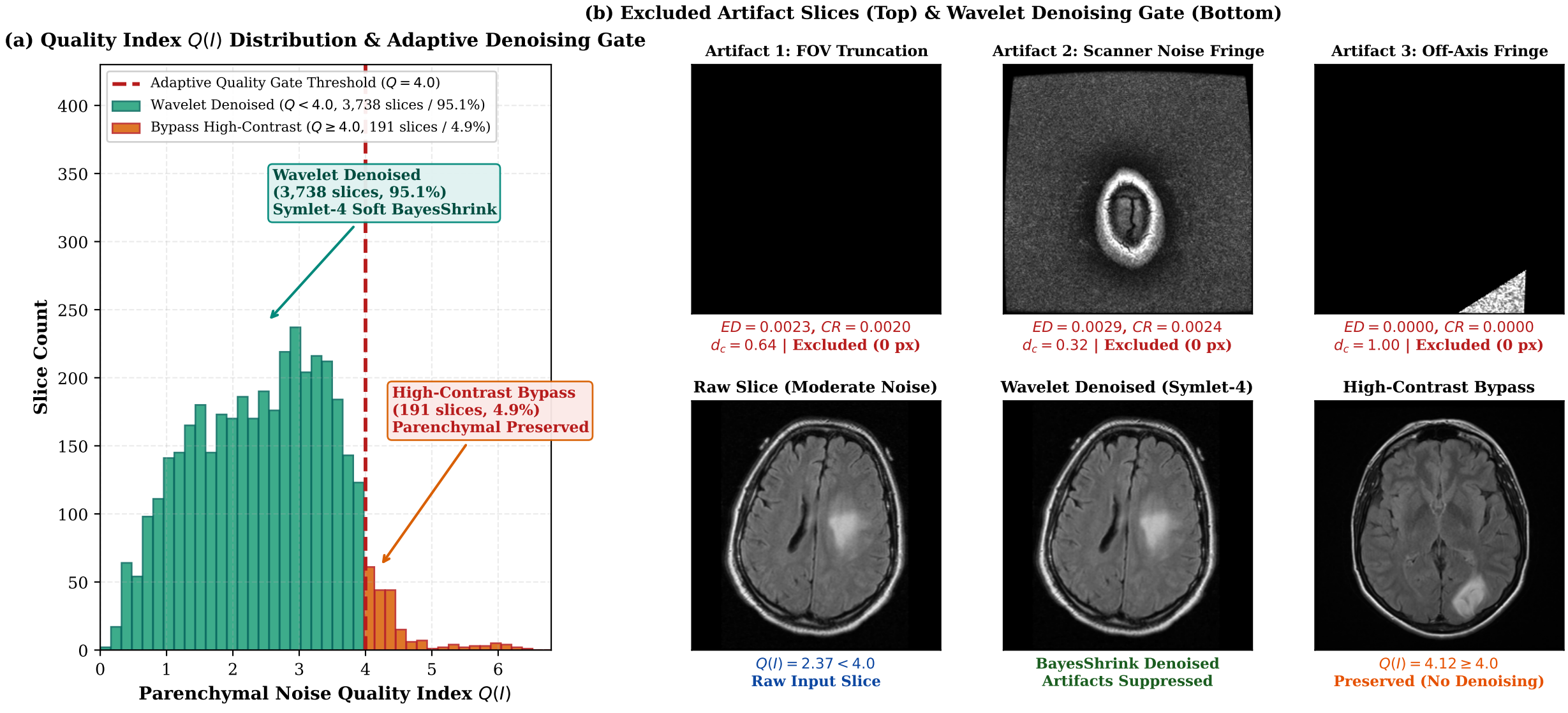

(a) Quality Index $Q(I)$ Distribution & Adaptive Denoising Gate
Adaptive Quality Gate Threshold ($Q = 4.0$)
Wavelet Denoised ($Q < 4.0$, 3,738 slices / 95.1%)
Bypass High-Contrast ($Q \geq 4.0$, 191 slices / 4.9%)
Wavelet Denoised
(3,738 slices, 95.1%)
Symlet-4 Soft BayesShrink
High-Contrast Bypass
(191 slices, 4.9%)
Parenchymal Preserved
Slice Count
Parenchymal Noise Quality Index $Q(I)$
(b) Excluded Artifact Slices (Top) & Wavelet Denoising Gate (Bottom)
Artifact 1: FOV Truncation
$ED = 0.0023$, $CR = 0.0020$
$d_c = 0.64$ | Excluded (0 px)
Artifact 2: Scanner Noise Fringe
$ED = 0.0029$, $CR = 0.0024$
$d_c = 0.32$ | Excluded (0 px)
Artifact 3: Off-Axis Fringe
$ED = 0.0000$, $CR = 0.0000$
$d_c = 1.00$ | Excluded (0 px)
Raw Slice (Moderate Noise)
$Q(I) = 2.37 < 4.0$
Raw Input Slice
Wavelet Denoised (Symlet-4)
BayesShrink Denoised
Artifacts Suppressed
High-Contrast Bypass
$Q(I) = 4.12 \geq 4.0$
Preserved (No Denoising)